Project Title

Fast Substring Search Using the Knuth–Morris–Pratt (KMP) Algorithm

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

String matching is a fundamental problem in computer science, with applications ranging from text editors to bioinformatics. Naive pattern-matching approaches can degrade to quadratic time on repetitive inputs, making them unsuitable for large-scale text search. This project presents an implementation and analysis of the Knuth–Morris–Pratt (KMP) algorithm, which guarantees linear-time string matching by avoiding redundant character comparisons. The core of the algorithm is a precomputed prefix (failure) function that captures self-overlaps within the pattern, allowing the search pointer to skip ahead intelligently after a mismatch instead of restarting. The project covers the construction of the prefix table, the linear-time scanning phase, and a complexity analysis showing the $O(n + m)$ bound. A working implementation is benchmarked against naive search on large text files to demonstrate the practical speedup, particularly on inputs with repetitive structure. The project concludes by discussing extensions such as multi-pattern matching and comparisons with alternative algorithms like the Z-algorithm.

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